

Going On the Offensive with Data Science in Infrastructure Planning

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Introduction

The Past and the Future

By necessity, infrastructure planning methodologies in the past have been defensive and labor-intensive, due to lack of insight into key datapoints. However, as the number of services we host continues to grow, we must evolve beyond cluster micromanagement and manual toll-keeping of the cloud.

Dramatic improvements in our understanding of cluster and workload behaviors, gained as part of our ongoing data enrichment efforts, have allowed us to produce new modeling with exactly the type of **predictive analytics** we need in order to move from “watch and see” to “automate and optimize”.

Automated and Data-Driven

The new planning model we’re introducing is an automated, data-driven process that is rooted in higher order mathematics, keying off of the aspects of our data proven to be the strongest predictors of future behavior.

The underlying calculations leverage data features including workload maturity classifiers, the service each cluster’s workloads provide, the region in which each cluster operates, and corresponding mortality rate metrics in order to forecast growth.

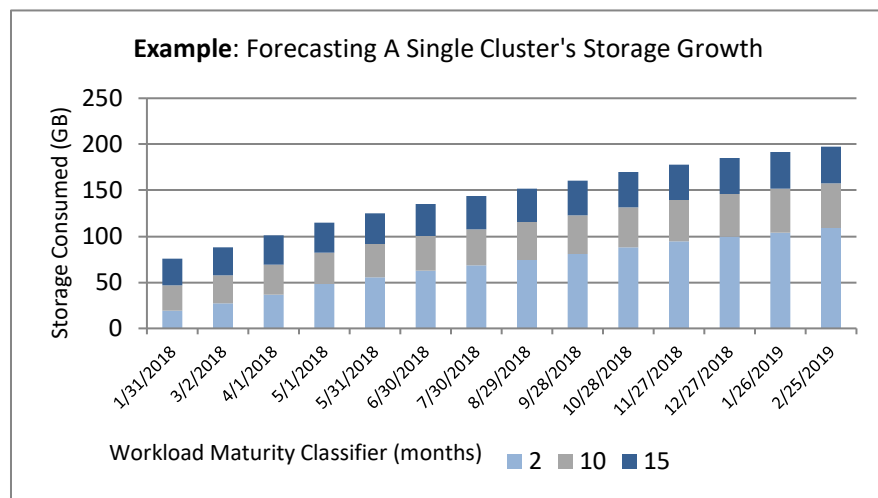


Figure 1: Example of storage growth prediction

IPCS Predictive Model

Workload Maturity and Growth Rate Dependency

Fundamental idea behind our methodology is dependency between workload maturity and its growth rate. We noticed that the rate the workloads grow depends on how old they are. By grouping workloads by their maturity level and service type, we were able to calculate 30 days growth rates for each group

and further (= as a next step) calculate correlation coefficients between workload maturity and its growth rate (for each maturity group). After verifying existence of correlation (between workload maturity and its growth rate), we were able to predict future grow for each cluster on the basis of /based on its service type and maturity level of the workloads residing on a given cluster.

Since some workloads drop from clusters, we also calculated mortality rates for each group and applied it in our calculations.

Abovementioned growth rate for each workload was calculated according to the following formula:

$$GR(VM) = \frac{Committed\ GB\ (VM, D + 30) - Committed\ GB\ (VM, D)}{Committed\ GB\ (VM, D)}$$

Equation 1: Growth Rate Formula

where:

- $GR(VM)$ is Growth Rate of workload VM
- $Committed\ GB\ (VM, D+30)$ is storage consumed in GB for workload VM on a given day $D + 30$
- $D + 30$ denotes a day 30 days after D
- $Committed\ GB\ (VM, D)$ is storage consumed in GB for workload VM on a given day D

The graph below demonstrates the strong correlation between the workload maturity data feature and growth behavior, even though the correlation is non-linear.

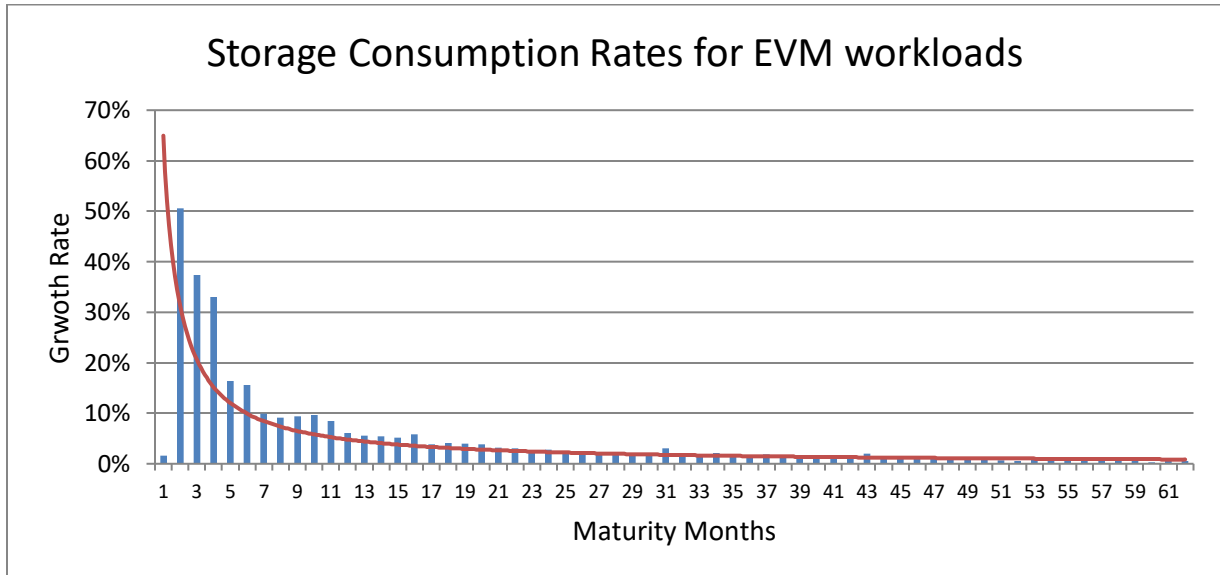


Figure 2: 30 days Growth Rate and Maturity Group correlation

As we can see, the data is not normally distributed and there is no linear dependency (straight line) between growth rate and maturity months for workloads, but there is definitely dependency (just not linear). In this case, we used non-parametric correlation including Spearman and Kendall rank-based correlation test.

Correlation Coefficients

The most commonly used correlation method is **Pearson Correlation** (correlation coefficient r). It measures a linear dependence between two variables, but it can only be used when those variables are from a normal distribution and their plot ($y = f(x)$ for variables x and y) is a linear regression curve.

For non-linear dependencies **Kendall tau** (correlation coefficient τ) and **Spearman rho** (correlation coefficient ρ) are most commonly used rank-based correlation coefficients (non-parametric).

The **Spearman correlation** method computes the correlation between the rank of first and the rank of the second variable.

The **Kendall correlation** method measures the correspondence between the rankings of two variables.

Pearson correlation formula

$$r = \frac{\sum_{k=1}^n (GR_k - \text{mean}(GR)) * (MG_k - \text{mean}(MG))}{\sqrt{\sum_{k=1}^n (GR_k - \text{mean}(GR))^2 * \sum_{k=1}^n (MG_k - \text{mean}(MG))^2}}$$

Equation 2: Pearson Correlation formula

where:

- GR is 30 days Growth Rate for a given Maturity Group (MG) of a specific workload type (vector of length n)
- MG is Maturity Group (in months) of a specific workload type (vector of length n)

Spearman correlation formula

$$\rho = \frac{\sum_{k=1}^n (GR'_k - \text{mean}(GR')) * (MG'_k - \text{mean}(MG'))}{\sqrt{\sum_{k=1}^n (GR'_k - \text{mean}(GR'))^2 * \sum_{k=1}^n (MG'_k - \text{mean}(MG'))^2}}$$

Equation 3: Spearman Correlation formula

where:

- $GR' = \text{rank}(GR)$
- $MG' = \text{rank}(MG)$

Kendall correlation formula

$$\tau = \frac{(\text{number of concordant pairs}) - (\text{number of discordant pairs})}{n * (n - 1)/2}$$

Equation 4: Kendall Correlation formula

where:

- *concordant pairs* are pairs of variables values growing or descending at the same time (when $GR_k < GR_j$ and $MG_k < MG_j$ or $GR_k > GR_j$ and $MG_k > MG_j$)
- *discordant pairs* are pairs of variables values not growing nor descending at the same time (when $GR_k < GR_j$ and $MG_k > MG_j$ or $GR_k > GR_j$ and $MG_k < MG_j$ - one grows and the other descends and the other way around)

Below we present results of correlation coefficient tests. Each coefficient (r , ρ , τ) implies that the correlation is significant (p-value is $< 5\%$), however, based on our data characteristics (non linear dependency), only rank- based coefficients should be taken into consideration.

Pearson correlation (r)

Table 1: Pearson correlation coefficients

Cluster type	p-value	p-value < 0.05 ?	Estimate
EVM	0.00000030	TRUE	-59.69%
MVM	0.00000548	TRUE	-54.16%
OPDB	0.00000213	TRUE	-56.08%
SSDC	0.00000639	TRUE	-53.83%

Spearman correlation (ρ)

Table 2: Spearman correlation coefficients

Cluster type	p-value	p-value < 0.05 ?	Estimate
EVM	0.00000000	TRUE	-86.78%
MVM	0.00000006	TRUE	-47.18%
OPDB	0.00000000	TRUE	-79.27%
SSDC	0.00000000	TRUE	-85.40%

Kendall correlation (τ)

Table 3: Kendall correlation coefficients

Cluster type	p-value	p-value < 0.05 ?	Estimate
EVM	0.00000000	TRUE	-94.65%
MVM	0.00000000	TRUE	-66.81%
OPDB	0.00000000	TRUE	-86.96%
SSDC	0.00000000	TRUE	-95.30%

As shown (displayed/illustrated) in the tables above, all of the workload types display high correlation between workload Growth Rate and its Maturity Group. This gives us solid base for using growth rates as determinant in our predictions.

Storage Consumption Prediction

Methodology

After establishing strong dependency between workload Growth Rate and Maturity Group (in months), we can apply it to predict storage consumption for a given cluster (and by this for a given workload type, pod, datacenter and region).

Presented below is the methodology used to create the storage consumption prediction.

For each maturity group in the given cluster, we calculated storage consumption forecast for the next month (based on the maturity distribution in a given time period on a given cluster).

$$\text{Committed GB}(D + 30, MG) = \text{Committed GB}(D, MG, WT) * (1 + GR(MG)) * (1 - MR(MG))$$

Equation 5: Storage consumption forecast formula

where:

- $\text{Committed GB}(D+30, MG)$ is storage consumption for a given maturity group (MG) of a specific workload type in day $D + 30$ (30 days after day D)
- $\text{Committed GB}(D, MG)$ is storage consumption for a given maturity group (MG) of a specific workload type on day D
- $GR(MG)$ is 30 days Growth Rate for a given maturity group (MG) of a specific workload type
- $MR(MG)$ is 30 days Mortality Rate for a given maturity group (MG) of a specific workload type

Algorithm

Procedure for Forecasting Storage Consumption based on Workload Maturity & Growth Rates

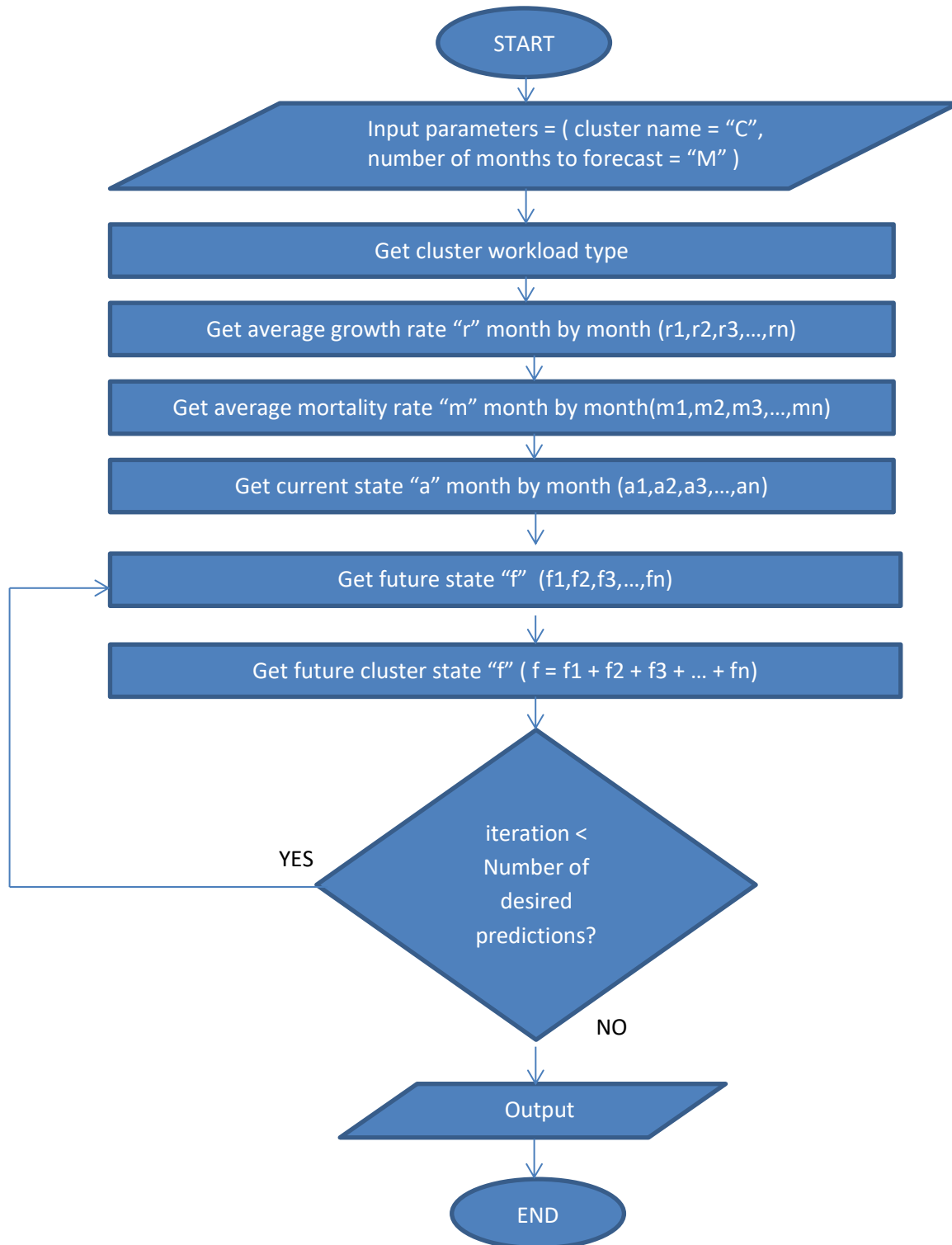
(Tommy's Procedure:)

1. Input parameters = (cluster name = "C", number of months to forecast = "M")
 - In this example we're going to use 12 months as the input M
 - In this example, we're going to use Muthu's cluster as the input C
2. Get cluster workload type from ccr_feed.dbo.vw_ccr_feed_with_otis_and_thresholds
 - In this example we are going to assume that the workload type is SSDC
3. Get average growth rate "r", month by month, for workloads of the same type. This should give you data like:
 - r1 = average 1 month growth rate for all SSDC workloads on all clusters in all regions
 - r2 = average 2 month growth rate for all SSDC workloads on all clusters in all regions
 - r3 = average 3 month growth rate for all SSDC workloads on all clusters in all regions
 - ...
 - r60 = average 60 month growth rate for all SSDC workloads on all clusters in all regions
4. Get average mortality rate "m", month by month, for workloads of the same type. This should give you data like:
 - m1 = average 1 month growth rate for all SSDC workloads on all clusters in all regions
 - m2 = average 2 month growth rate for all SSDC workloads on all clusters in all regions
 - m3 = average 3 month growth rate for all SSDC workloads on all clusters in all regions
 - ...
 - r60 = average 60 month growth rate for all SSDC workloads on all clusters in all regions
5. Get average provisioning rate "p", month by month, for workloads of the same type. -> future feature
6. Get current state "a" of the cluster storage utilization (by workload maturity distribution). This should give you data like:
 - a1 = sum of all gigabytes utilized by workloads of 1 month age, running on cluster C
 - a2 = sum of all gigabytes utilized by workloads of 2 months age
 - a3 = sum of all gigabytes utilized by workloads of 3 months age
 - ...
 - a60 = sum of all gigabytes utilized by workloads of 60 months age
7. For each maturity month get future state f
 - $f1 = a1 * (1+r1) * (1-m1) * p$
 - $f2 = a2 * (1+r2) * (1-m2)$
 - $f3 = a3 * (1+r3) * (1-m3)$
 - ...
 - $fn = an * (1+rn) * (1-mn)$
8. Get future state "f" of the cluster (storage utilization)
 - $f = f1 + f2 + f3 + \dots + fn$
9. For each month in the next M months (m1, m2, m3 ... m12)
 - f1 (Forecasted utilization for m1) = $\text{sum}(a1*r1, a2*r2, a3*r3, \dots a60*r60)$
 - f2 (Forecasted utilization for m2) = $\text{sum}(a1*r2, a2*r3, a3*r4 \dots a59*r60)$
 - f3 (Forecasted utilization for m3) = $\text{sum}(a1*r3, a2*r4, a3*r5 \dots a58*r60)$

- ...f12 = ...

10. Repeat step 7 and 8 for the number of desired predictions using a previous state as a new current state.

Diagram 1: Tommy's algorithm



MY PROCEDURE

1. For certain cluster CL get its type CL_TYPE and the initial state (IS) which consist of the number of workloads for each maturity group and sum of committed GB per each maturity group
Initial state IS = $f((\text{Maturity_Goup_1}, \text{Committed_GB_1}), (\text{Maturity_Goup_2}, \text{Committed_GB_2}), \dots, (\text{Maturity_Goup_n}, \text{Committed_GB_n}))$

$A(n)$

2. For each maturity group n find their growth rate $R(n)$ and mortality rate $M(n)$ specific for the cluster type (also cluster type provisioning prediction – future feature).
3. For each workload maturity group n follow formula

$$F(n,1) = S(n) * (1+R(n)) * (1-M(n))$$

where:

- $S(n)$ is sum of committed GB for Maturity Group n
- $R(n)$ is Growth Rate for Maturity Group n
- $M(n)$ is Mortality Rate for Maturity Group n

4. Save output of step 3 as new initial state ($F(n,2)$)
5. Repeat step 3 m times to obtain m predictions ($F(n,m)$)
6. For each prediction step add all of Committed GB

$$F(1) = F(1,1) + F(2,1) + F(3,1) + \dots + F(n,1)$$

$$F(2) = F(1,2) + F(2,2) + F(3,2) + \dots + F(n,2)$$

$$F(3) = F(1,3) + F(2,3) + F(3,3) + \dots + F(n,3)$$

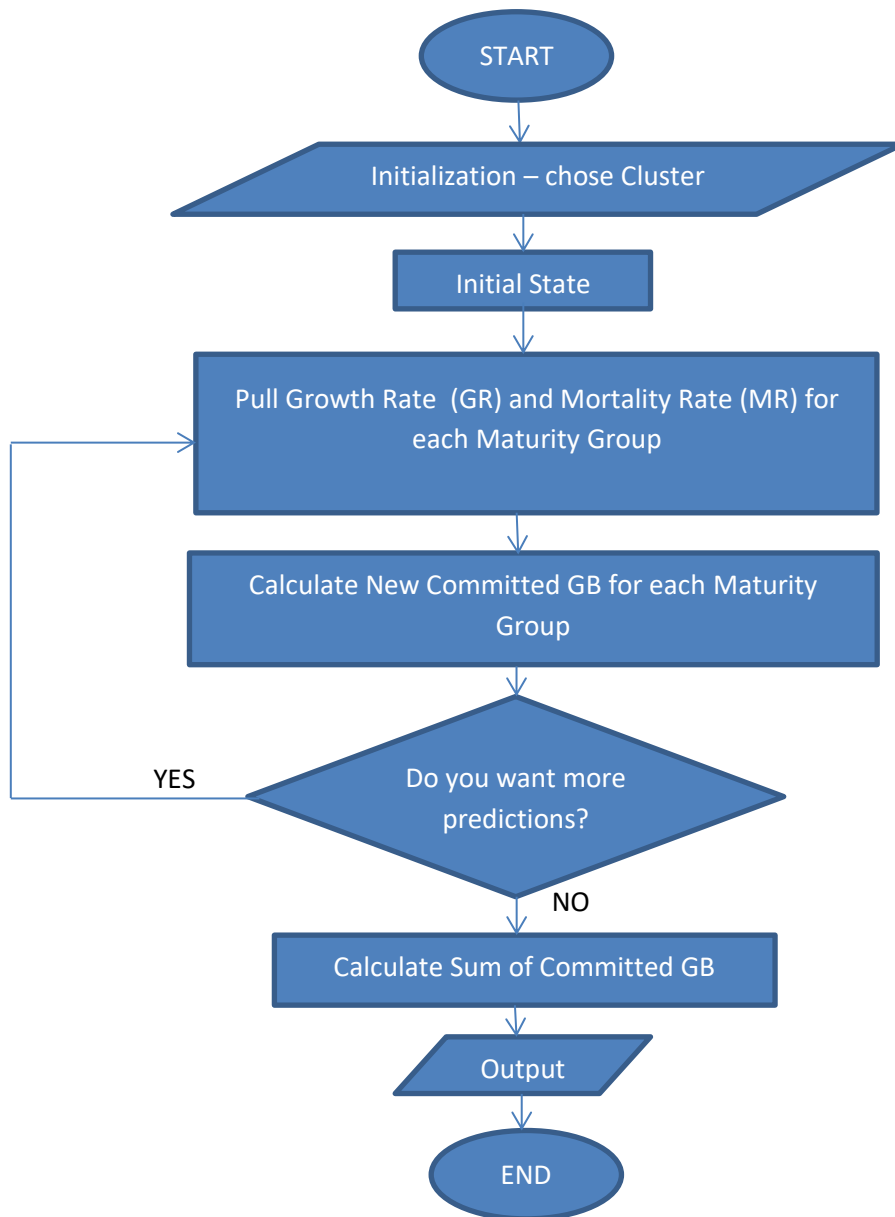
...

$$F(m) = F(1,m) + F(2,m) + F(3,m) + \dots + F(n,m)$$

Where:

- $F(1)$ is predicted storage consumed in GB 30 days after the initial state.
- $F(2)$ is predicted storage consumed in GB 30 days after $F(1)$ (so 60 days after initial state).
- ...
- $F(m)$ is predicted storage consumed in GB $m * 30$ days after initial state.

Diagram 2: Marta's algorithm



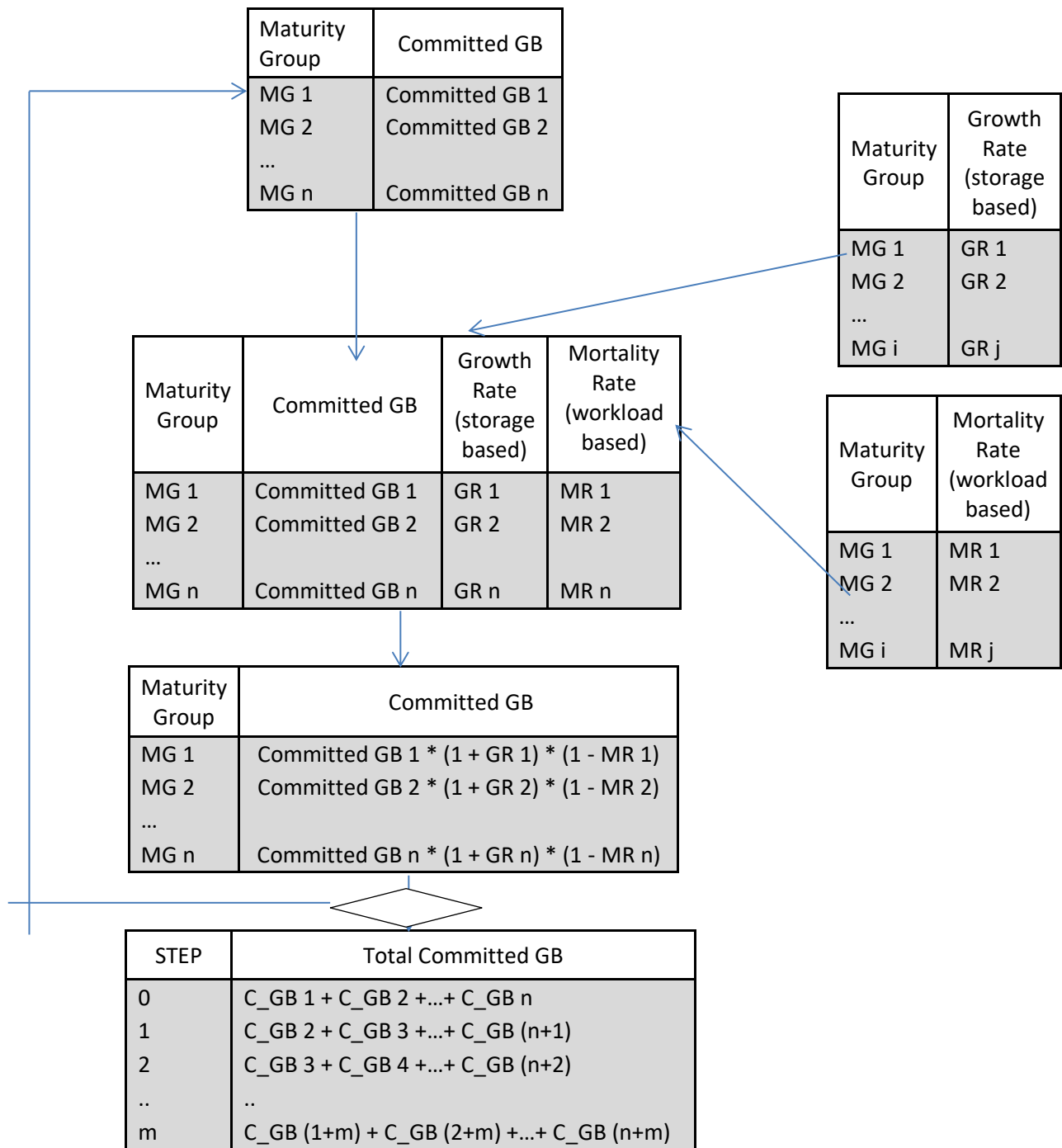
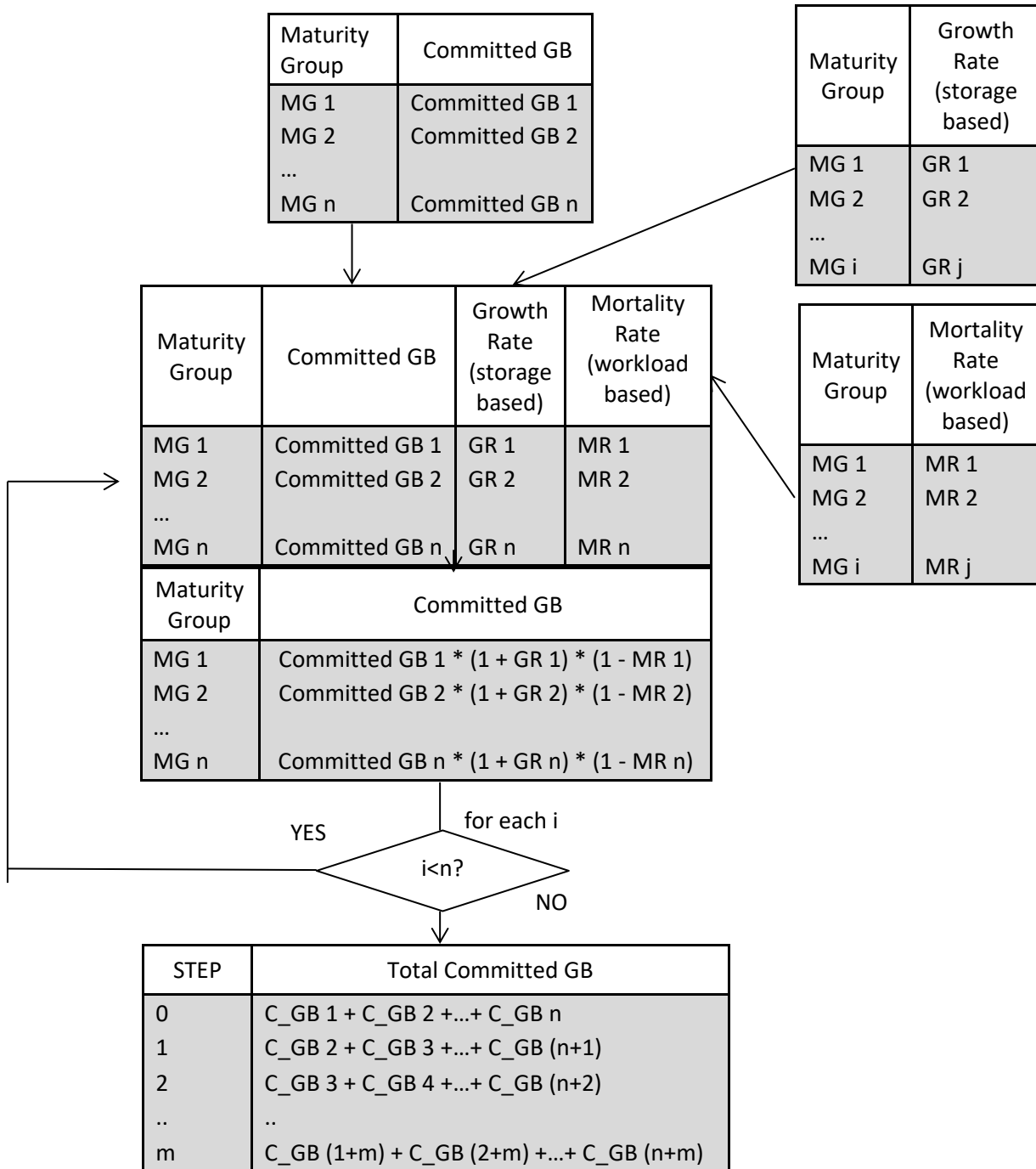


Diagram 3: Detailed algorithm description



Example

Our methodology can be best described using an example.

For the simplicity and clarity of the example, we're using a cluster with only 3 groups of maturity level.

Let's take into consideration EVM cluster [REDACTED]

Presented chart depicts Maturity Group distribution on that cluster on 2018/01/31.

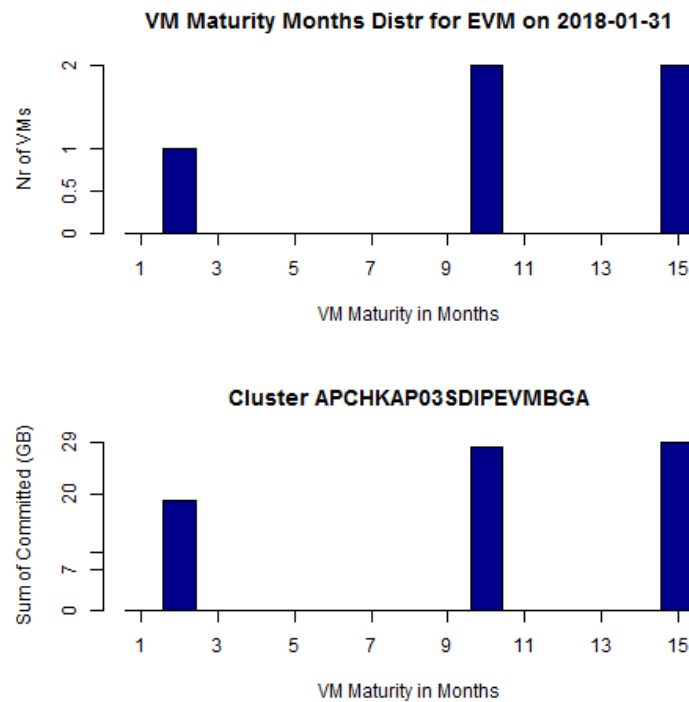


Figure 3: Workload Maturity distribution in a cluster

We can see that on that day, cluster consisted of 5 workloads total – 1 workload that was 2 months old, 2 workloads that were 10 months old and 2 workloads that were 15 months old.

Next step is to match Growth Rates and Mortality Rates for EVM type workloads for Maturity Group 2, 10 and 15. We apply Growth Rates and Mortality Rates to the formula (Equation 5: Storage consumption forecast formula) and obtain storage consumption prediction for the day 30 days after our start date ($1/31/2018 + 30 = 3/2/2018$). Our prediction indicates storage in 30 days because we used 30 days Growth Rates. If we used 90 days Growth Rates, then we would predict storage in 90 days. Next step is repeating step 1 for the new data. Notice that after 30 days, we need to match growth and mortality rates for Maturity Group 3, 11 and 16 etc. Finally, we add up all the predicted storage for a given day to obtain the prediction.

Table 4: Storage Consumption Prediction - cluster APCHKAP03SDIPEVMBGA

Date	Maturity Group	Workload Count	Growth Rate (storage based)	Mortality Rate (workload based)	Committed GB	Total Committed GB
1/31/2018	2	1	50.59%	4.52%	19.00	76.00
1/31/2018	10	2	9.61%	0.58%	28.00	
1/31/2018	15	2	5.15%	0.95%	29.00	
3/2/2018	3	1	37.37%	2.48%	27.32	88.04
3/2/2018	11	2	8.44%	1.09%	30.51	
3/2/2018	16	2	5.85%	1.19%	30.20	
4/1/2018	4	1	32.99%	1.08%	36.60	100.92
4/1/2018	12	2	6.09%	0.78%	32.73	
4/1/2018	17	2	3.85%	0.96%	31.59	
5/1/2018	5	1	16.44%	1.19%	48.15	115.10
5/1/2018	13	2	5.52%	0.69%	34.46	
5/1/2018	18	2	4.17%	0.88%	32.49	
5/31/2018	6	1	15.57%	1.62%	55.40	125.06
5/31/2018	14	2	5.42%	0.85%	36.11	
5/31/2018	19	2	3.92%	0.88%	33.55	
6/30/2018	7	1	9.93%	0.92%	62.99	135.29
6/30/2018	15	2	5.15%	0.95%	37.74	
6/30/2018	20	2	3.87%	0.80%	34.56	

Graph below (Figure 4: Detailed Storage Consumption prediction for EVM cluster APCHKAP03SDIPEVMBGA) represents numbers from the table above (Table 4: Storage Consumption Prediction - cluster APCHKAP03SDIPEVMBGA).

As we can see, the youngest workload group grows the fastest. You can see it in the table above (growth rates) as well as you can see the light blue bars on the picture below growing faster than other bars. It means that younger workloads grow faster. This also depicts difference in growth rates depending on the maturity group level (younger workload grow faster than older once).

Last graph showing our cluster (Figure 5: Storage Consumption prediction for EVM cluster XXXXXXXXXX) is our final prediction. We can see historical data (dark blue) as well as new predicted storage consumed in GB (light blue).

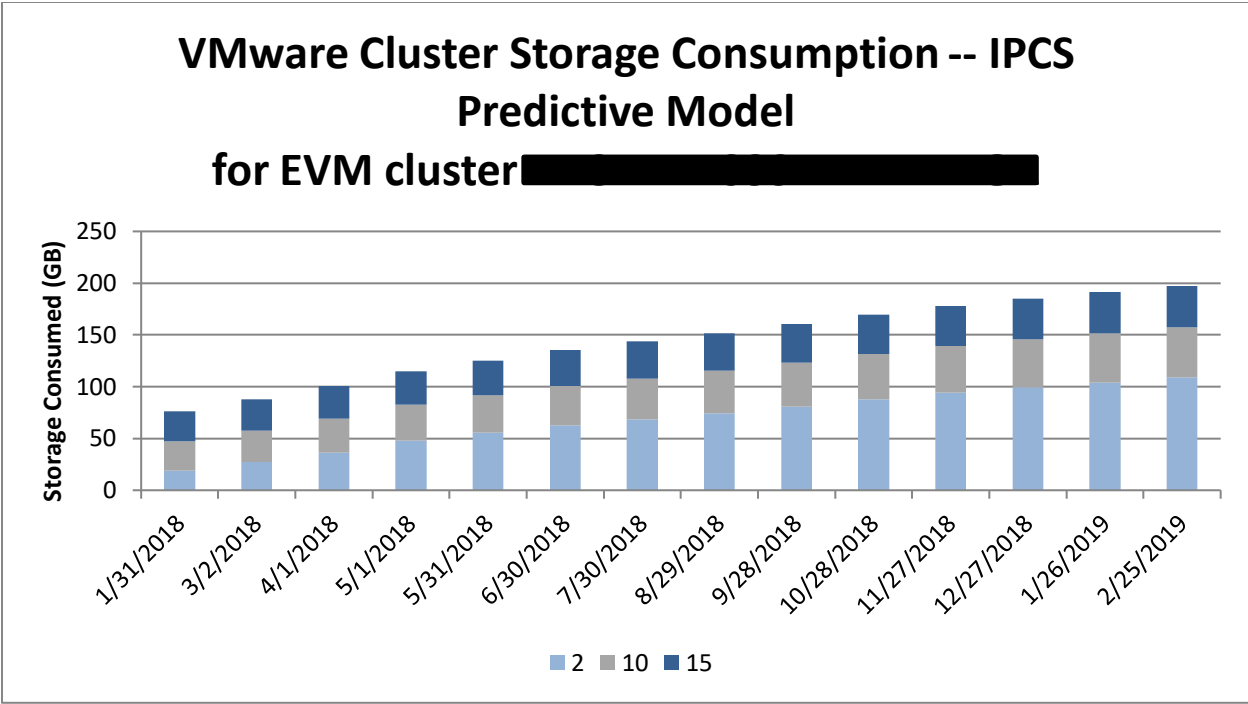


Figure 4: Detailed Storage Consumption prediction for EVM cluster [REDACTED]

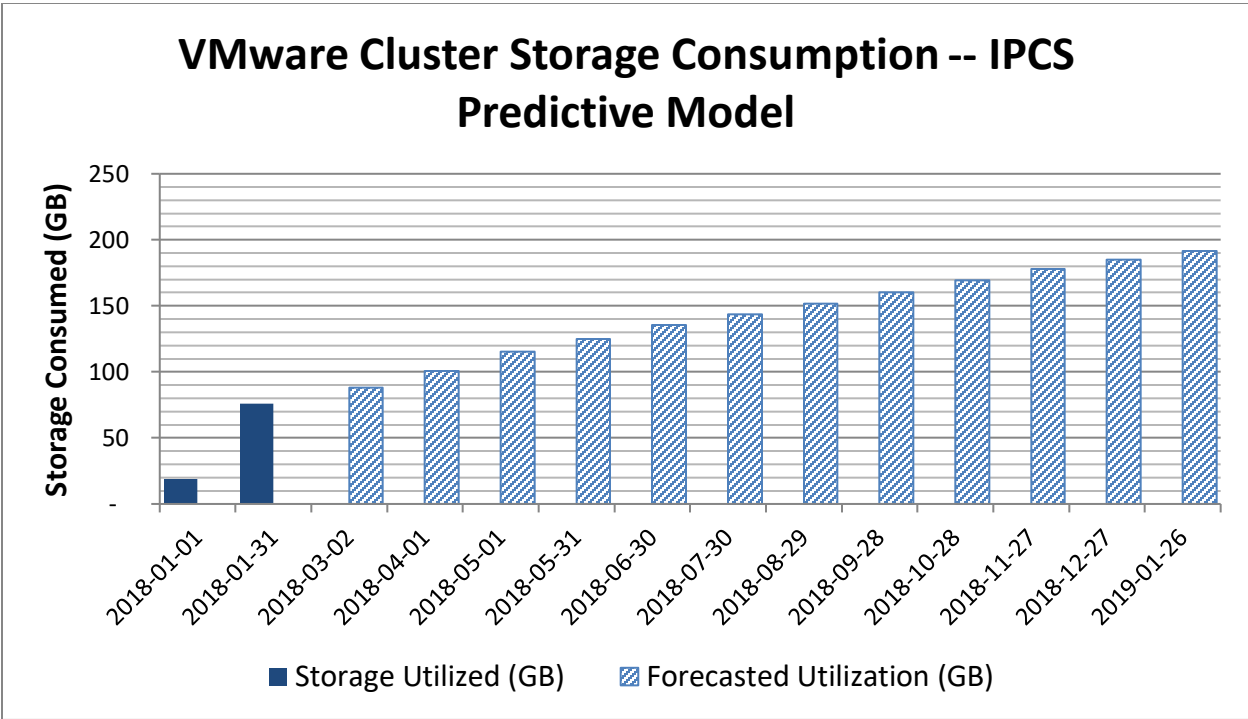


Figure 5: Storage Consumption prediction for EVM cluster [REDACTED]

Given example reveals one of the simplest workload maturity distribution within a cluster.

In many cases the older the cluster, the more complicated the workload maturity distribution on it.

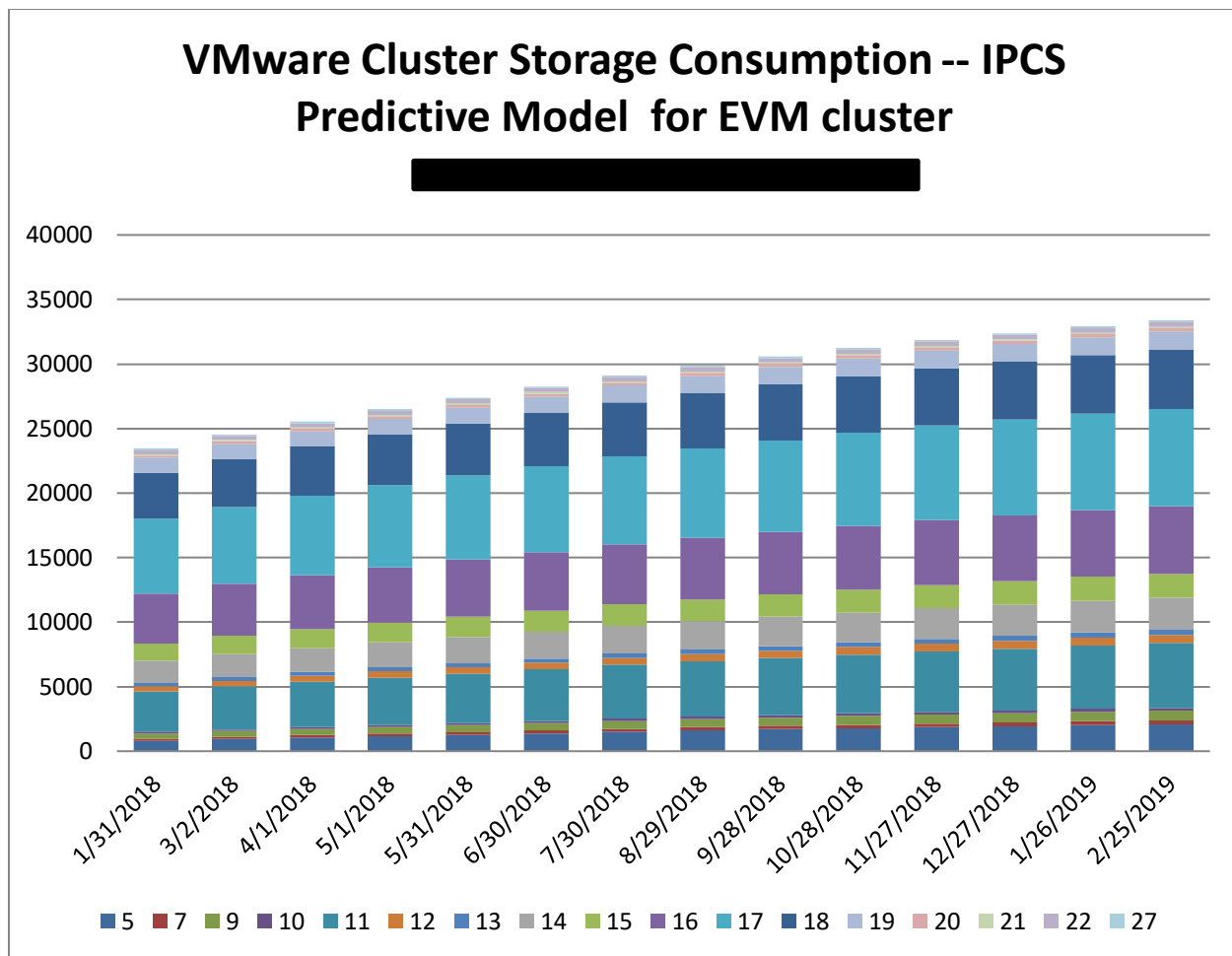


Figure 6: Detailed Storage Consumption Prediction - EVM cluster

As illustrated on the graph (Figure 6: Detailed Storage Consumption Prediction - EVM cluster), EVM cluster on 1/31/2018 consisted of 137 workloads which can be group into 17 maturity groups all the way from 5 months to 27 months.

As can be seen on the chart, they didn't always grow in the same rate.

This example is more common among our clusters than the previous one.

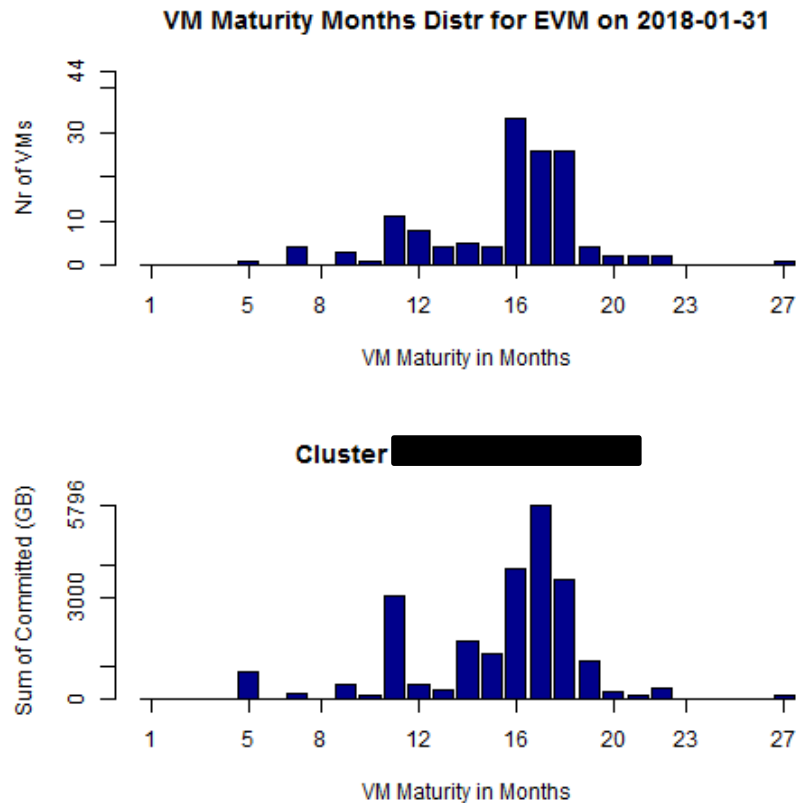


Figure 7: Workload Maturity Group Distribution

Cumulative Predictions

As explained, we are able to predict storage consumed in GB per each cluster.

Having those predictions, we can always cumulate them to examine the future state of storage in a bigger scope (region, data center, pod, cluster type).

Graph below displays historical data and future prediction of storage consumed for all the EVM clusters.

As we can see our model is immune to temporary disturbances in the data. This is because we're not directly using historical data as a time series for the prediction.

The disturbances in the graph are due to moving backup from Net Backup to object storage.

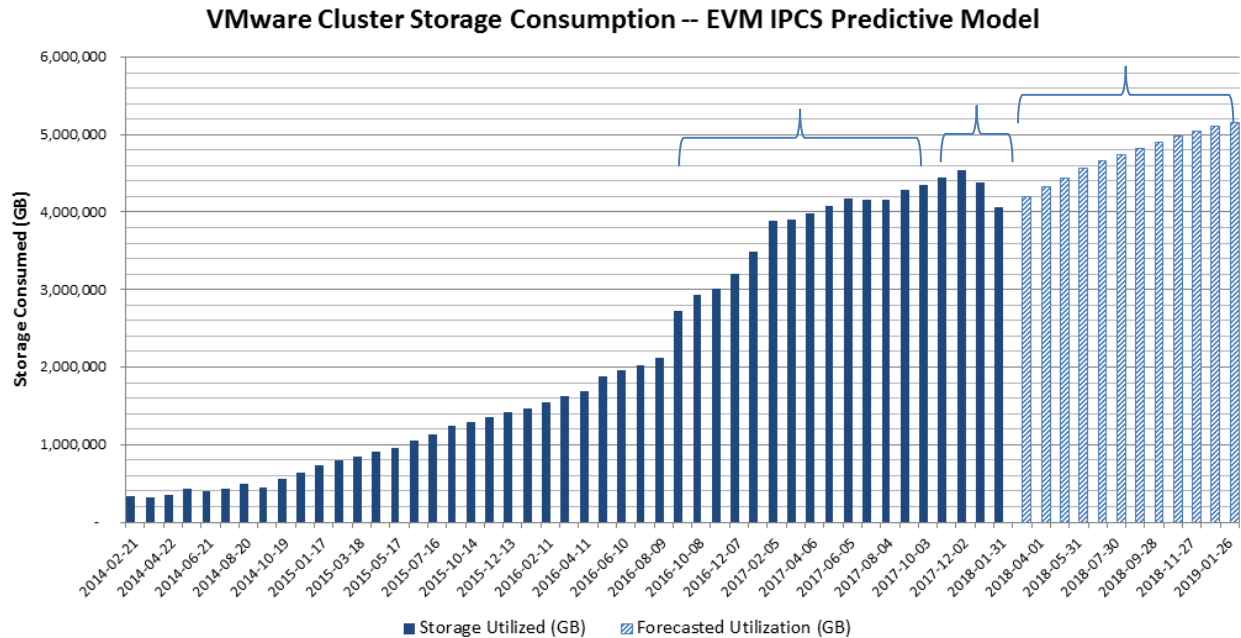


Figure 8: Storage Consumption Prediction for EVM clusters

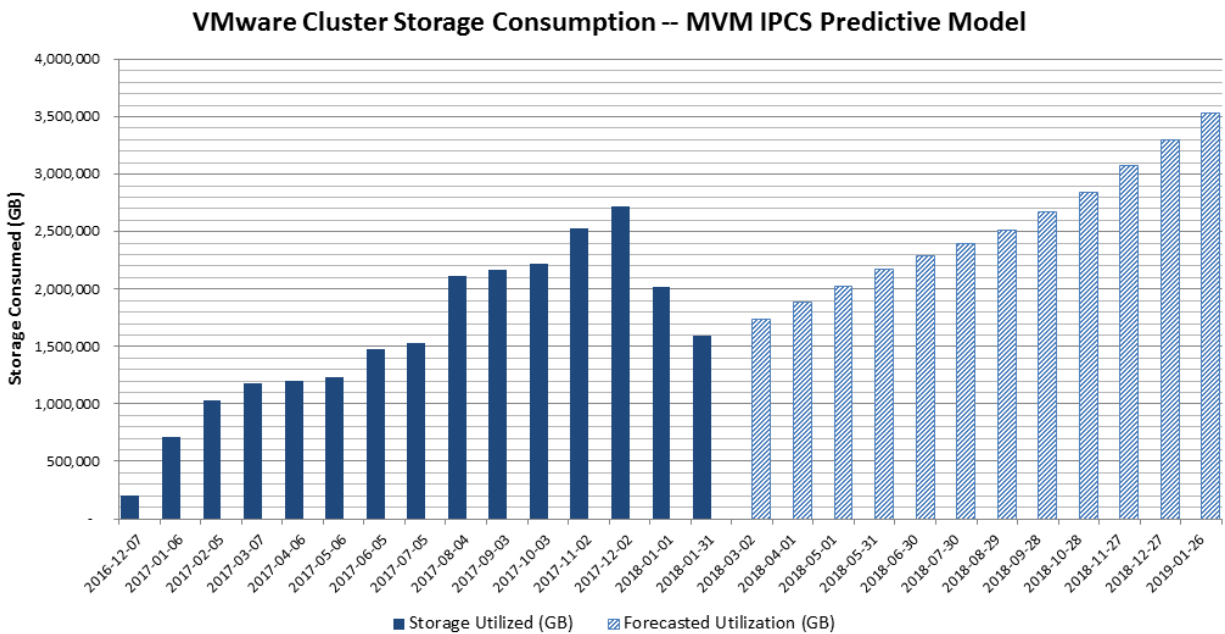


Figure 9: Storage Consumption Prediction for MVM clusters

Graph above (Figure 9: Storage Consumption Prediction for MVM clusters) exposes similar disturbances as graph for EVM clusters Storage Consumption Prediction (Figure 8: Storage Consumption Prediction for EVM clusters). As we can see, the growth here is steeper/ more aggressive and there's more

inconsistency among the data. MVM clusters are much younger and have different characteristics than EVM clusters. Nevertheless, the prediction displays expected growth. (nothing surprising, the prediction looks like we would expect it to look)

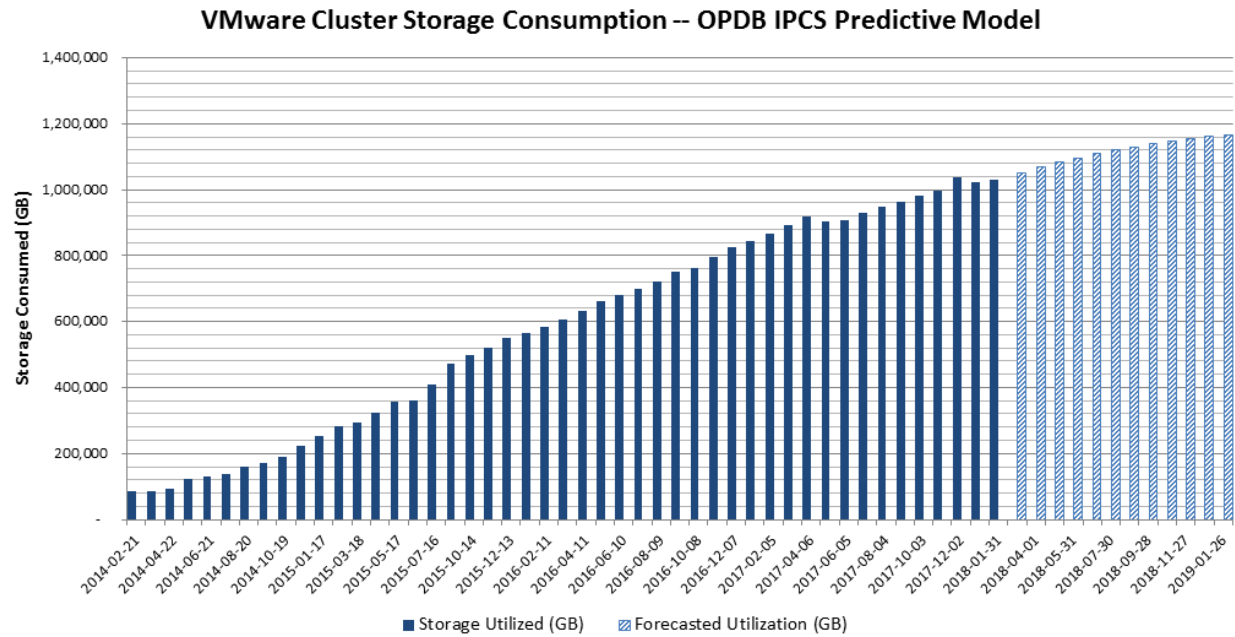


Figure 10: Storage Consumption Prediction for OPDB clusters

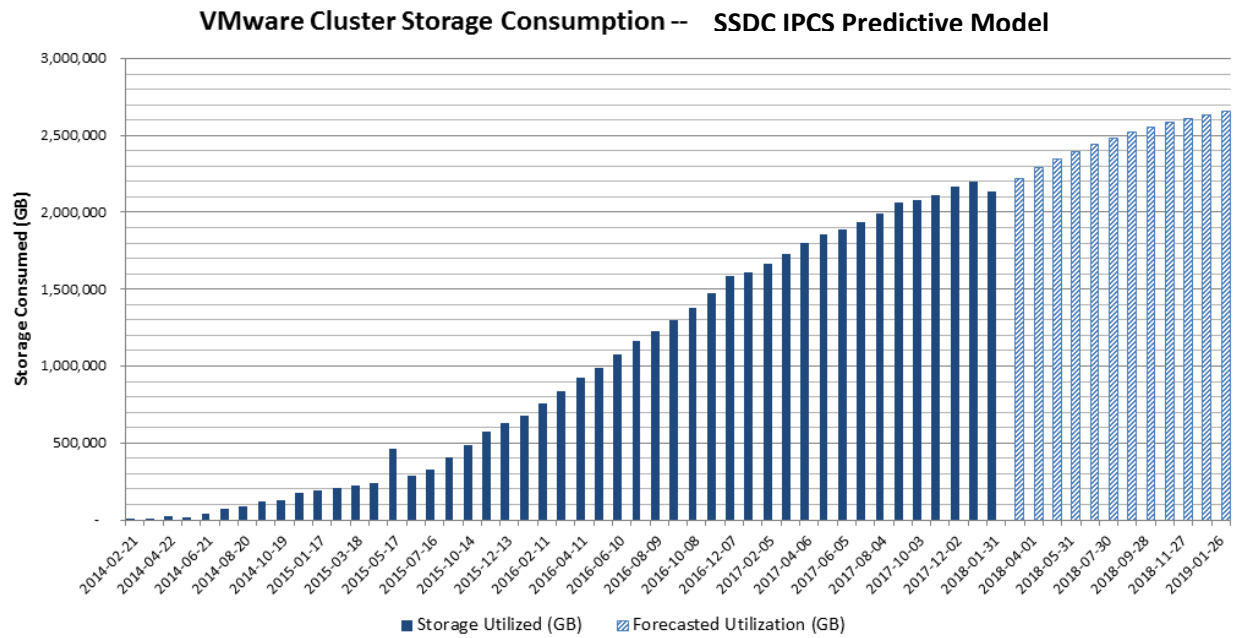


Figure 11: Storage Consumption Prediction for SSDC clusters

In contrary to EVM and MVM clusters, storage consumed for OPDB and SSDC clusters growths pretty smoothly, without much volatility.

By looking at single clusters, we can differentiate the aggressiveness of the growth depending on the number and age of the workloads residing on the cluster.

For example, EVM cluster [REDACTED]

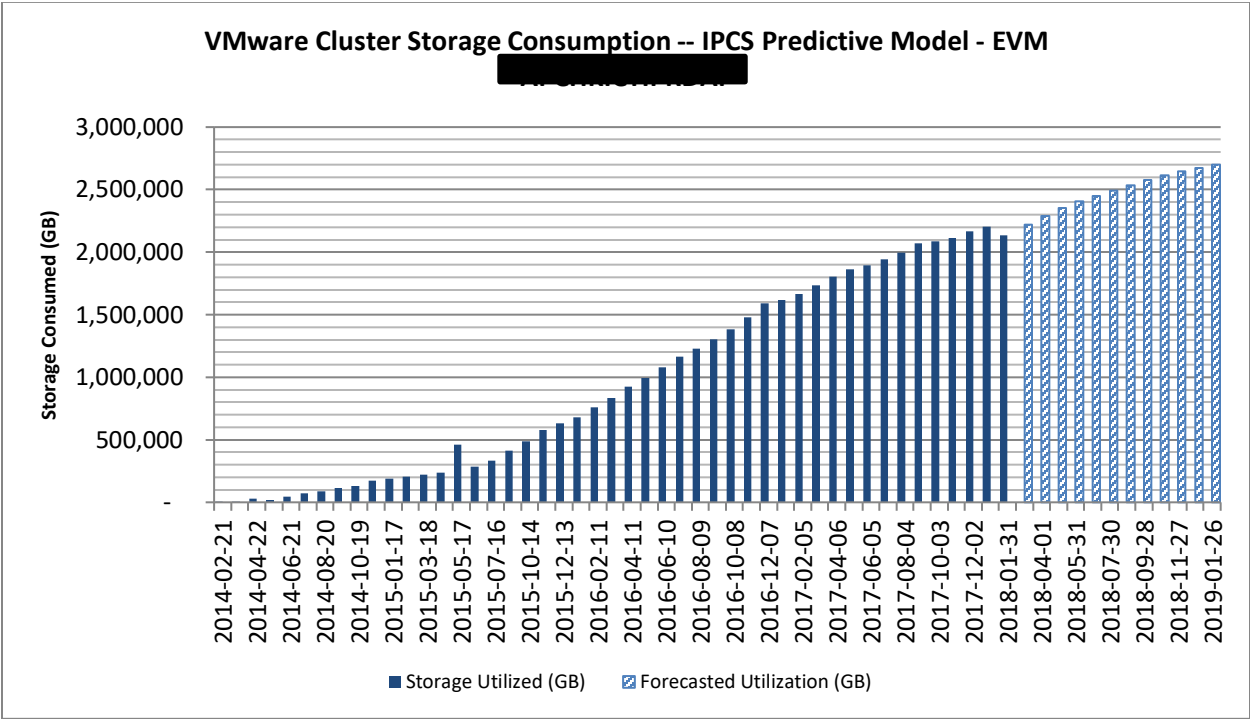


Figure 12: Storage Consumption Prediction for old cluster

Compared to [REDACTED]

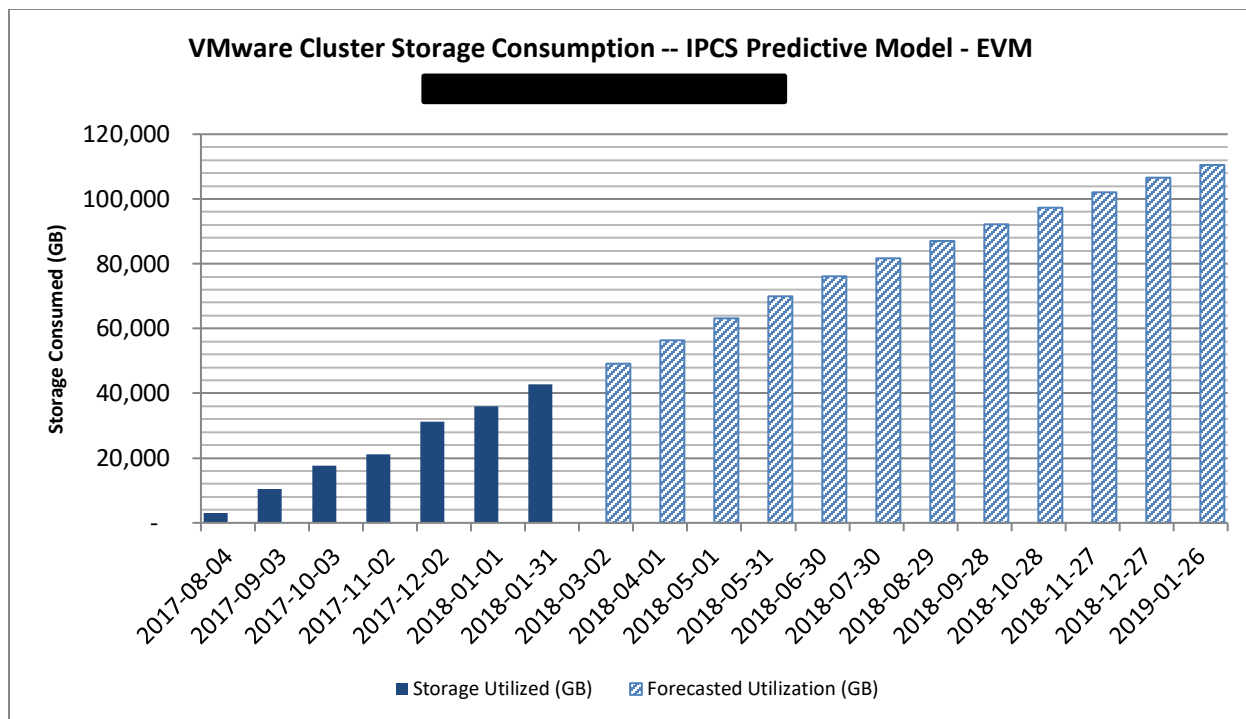


Figure 13: Storage Consumption Prediction for young clusters

Conclusion

Provided analysis and its results gives us solid foundation to use workload maturity with workload type as a determinant to predict storage consumption for a given cluster, pod, data center, region or workload type.

This information can lead to a more precise prediction which can be immune to a local disturbances in the data (for example resulting from moving storage from one cluster to the other).

Presented model is just the basic model that can be enriched with more functionality.

Next steps in developing our model would be use of provisioning rates and its applications (depending on the request). Knowing when to use it and when not

After reaching those goals, we can expand the model to more complex model predicting /taking into consideration also memory and CPU.

Further steps include predicting number of workloads that can be accommodated on a certain cluster.